

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12408650
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Pattison; William J. et al.

Heat system for killing pests

Abstract

The present disclosure generally relates to the field of pest elimination including all life stages of bed bugs. The present disclosure includes articles, systems, and methods of heat treatment to target and kill pests.

Inventors: Pattison; William J. (Greensboro, NC), Barcay; S. John (Burnsville, MN), Lang; Jason G. (Bloomington, MN), Wagner; Christopher C. (St. Paul, MN), Diller; Christopher D. (Bloomington, MN), Olson; Joelle F. (Hanover, MN)

Applicant: Ecolab USA Inc. (St. Paul, MN)

Family ID: 46827309

Assignee: ECOLAB USA INC. (St. Paul, MN)

Appl. No.: 18/769779

Filed: July 11, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20250008938 A1	Jan. 09, 2025

Related U.S. Application Data

continuation parent-doc US 18348754 20230707 US 12063921 child-doc US 18769779
continuation parent-doc US 17236187 20210421 US 11737445 20230829 child-doc US 18348754
continuation parent-doc US 16050016 20180731 US 11013226 20210525 child-doc US 17236187
division parent-doc US 14954352 20151130 US 10070639 20180911 child-doc US 16050016
division parent-doc US 13421409 20120315 US 9226489 20160105 child-doc US 14954352
us-provisional-application US 61454255 20110318

Publication Classification

Int. Cl.: A01M1/20 (20060101)

U.S. Cl.:

CPC A01M1/2094 (20130101);

Field of Classification Search

CPC: A01M (1/2094)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
51493	12/1864	Tapper	N/A	N/A
56008	12/1865	Clarke	N/A	N/A
57338	12/1865	Lake	N/A	N/A
137936	12/1872	Legab	N/A	N/A
139562	12/1872	Fewell	N/A	N/A
161032	12/1874	Hawkins	N/A	N/A
168306	12/1874	Watkins	N/A	N/A
317322	12/1884	Dosh	N/A	N/A
431815	12/1889	Linder et al.	N/A	N/A
481270	12/1891	Baggesen	N/A	N/A
616049	12/1897	Archer	N/A	N/A
844919	12/1906	Brundin	N/A	N/A
1614157	12/1926	Schneider	N/A	N/A
3056415	12/1961	Nimmo	N/A	N/A
3327441	12/1966	Kelly	N/A	N/A
3594481	12/1970	Lindberg et al.	N/A	N/A
3676486	12/1971	Nikles et al.	N/A	N/A
3714947	12/1972	Hardy	N/A	N/A
3878570	12/1974	Donnelly	N/A	N/A
3901437	12/1974	Harkins	N/A	N/A
4033367	12/1976	Johnston	N/A	N/A
4086424	12/1977	Mellen, Sr.	N/A	N/A
4228511	12/1979	Simcoe et al.	N/A	N/A
4305235	12/1980	Roston	N/A	N/A
4323582	12/1981	Siegel et al.	N/A	N/A
4327710	12/1981	DeLoach et al.	N/A	N/A
4416658	12/1982	Numazawa et al.	N/A	N/A
4524264	12/1984	Takeuchi et al.	N/A	N/A
4557317	12/1984	Harmon, Jr.	N/A	N/A
4702413	12/1986	Beckey et al.	N/A	N/A
4788149	12/1987	Cerami et al.	N/A	N/A
4817329	12/1988	Forbes	N/A	N/A
4843576	12/1988	Smith et al.	N/A	N/A

4941499	12/1989	Pelsue et al.	N/A	N/A
4953320	12/1989	Nelson	N/A	N/A
4958456	12/1989	Chaudoin	N/A	N/A
4961283	12/1989	Forbes	N/A	N/A
4966755	12/1989	Smith	N/A	N/A
5011909	12/1990	Borovsky et al.	N/A	N/A
5058313	12/1990	Tallon	N/A	N/A
5203108	12/1992	Washburn, Jr.	N/A	N/A
5244001	12/1992	Lynch	N/A	N/A
5282334	12/1993	Kimura	N/A	N/A
5349778	12/1993	Chu	N/A	N/A
5358934	12/1993	Borovsky et al.	N/A	N/A
5439821	12/1994	Borovksy et al.	N/A	N/A
5459130	12/1994	Borovsky et al.	N/A	N/A
5471782	12/1994	Brittell	N/A	N/A
5479743	12/1995	Queen et al.	N/A	N/A
5501976	12/1995	Borovsky et al.	N/A	N/A
5511254	12/1995	O'Brien	N/A	N/A
5555927	12/1995	Shah	N/A	N/A
5569401	12/1995	Gilliland et al.	N/A	N/A
5575106	12/1995	Martin et al.	N/A	N/A
5613320	12/1996	Thomasson et al.	N/A	N/A
5629196	12/1996	Borovsky et al.	N/A	N/A
5638848	12/1996	November	N/A	N/A
5641463	12/1996	Langhart	N/A	N/A
5658130	12/1996	Goldstein	N/A	N/A
5661637	12/1996	Villaume	N/A	N/A
5768907	12/1997	Lee	N/A	N/A
RE35834	12/1997	Miller	N/A	N/A
5792419	12/1997	Williamson et al.	N/A	N/A
5792750	12/1997	Borovsky et al.	N/A	N/A
D426596	12/1999	Polk	N/A	N/A
6141901	12/1999	Johnson et al.	N/A	N/A
6171561	12/2000	Williamson et al.	N/A	N/A
6192598	12/2000	Halverson et al.	N/A	N/A
6279261	12/2000	Binker et al.	N/A	N/A
6302094	12/2000	Wehrly	N/A	N/A
6313441	12/2000	Schaper et al.	N/A	N/A
6327812	12/2000	Hedman et al.	N/A	N/A
6337080	12/2001	Fryan et al.	N/A	N/A
6467215	12/2001	Nelson et al.	N/A	N/A
6510565	12/2002	Zvezdaryk	N/A	N/A
6568123	12/2002	Nelson et al.	N/A	N/A
6588140	12/2002	Johnson et al.	N/A	N/A
6612067	12/2002	Topp	N/A	N/A
6647661	12/2002	Grigorov	N/A	N/A
6678994	12/2003	Topp	N/A	N/A

6772829	12/2003	Lebrun	N/A	N/A
6827036	12/2003	Connolly	N/A	N/A
6901317	12/2004	Starner	N/A	N/A
7134239	12/2005	Barra	N/A	N/A
7168630	12/2006	Ketcha	N/A	N/A
7195025	12/2006	Choi et al.	N/A	N/A
7335733	12/2007	Paesen et al.	N/A	N/A
7410058	12/2007	Kirkegaard	N/A	N/A
7481234	12/2008	Gustafson	N/A	N/A
7591099	12/2008	Lang	N/A	N/A
7676985	12/2009	Perkins	N/A	N/A
7690148	12/2009	Hedman	N/A	N/A
7719429	12/2009	Barber	N/A	N/A
7743552	12/2009	Borth	N/A	N/A
7926222	12/2010	Molnar	N/A	N/A
9101125	12/2014	Knote	N/A	N/A
9226489	12/2015	Pattison	N/A	A01M 1/2094
9609857	12/2016	Nugent	N/A	N/A
10070639	12/2017	Pattison	N/A	A01M 1/2094
10477853	12/2018	Wheeler	N/A	F24H 3/02
11013226	12/2020	Pattison	N/A	A01M 1/2094
11737445	12/2022	Pattison	43/132.1	A01M 1/2094
12063921	12/2023	Pattison et al.	N/A	N/A
2001/0004813	12/2000	Hedman	N/A	N/A
2002/0011020	12/2001	Nelson	N/A	N/A
2002/0040027	12/2001	Karrer et al.	N/A	N/A
2002/0062852	12/2001	Jopp et al.	N/A	N/A
2002/0066223	12/2001	Hedman et al.	N/A	N/A
2003/0015225	12/2002	Nightingale	N/A	N/A
2003/0029605	12/2002	Lebrun	N/A	N/A
2003/0056390	12/2002	Adrian	N/A	N/A
2003/0078390	12/2002	Paesen et al.	N/A	N/A
2003/0170257	12/2002	Trimnell et al.	N/A	N/A
2004/0039431	12/2003	Machold et al.	N/A	N/A
2004/0231687	12/2003	Kirkegaard	N/A	N/A
2005/0000937	12/2004	Chiang et al.	N/A	N/A
2005/0086858	12/2004	Schmidt	N/A	N/A
2005/0123556	12/2004	Nuttall et al.	N/A	N/A
2005/0220662	12/2004	Hedman	N/A	N/A
2005/0224492	12/2004	Roy	N/A	N/A
2005/0246942	12/2004	Mueller et al.	N/A	N/A
2005/0268543	12/2004	Hicks et al.	N/A	N/A
2006/0057100	12/2005	Geier et al.	N/A	N/A
2006/0213546	12/2005	Mitsui et al.	N/A	N/A
2006/0269582	12/2005	Bruins et al.	N/A	N/A
2006/0277830	12/2005	Boggs, Jr.	N/A	N/A

2007/0016271	12/2006	Hammond	N/A	N/A
2007/0031411	12/2006	Trimnell et al.	N/A	N/A
2007/0044372	12/2006	Lang et al.	N/A	N/A
2007/0050903	12/2006	Sappenfield	N/A	N/A
2007/0283986	12/2006	Baum	N/A	N/A
2008/0014111	12/2007	Hedman	N/A	N/A
2008/0148624	12/2007	Borth et al.	N/A	N/A
2008/0221536	12/2007	Thomsen et al.	N/A	N/A
2008/0222808	12/2007	Bell et al.	N/A	N/A
2008/0248534	12/2007	Lim et al.	N/A	N/A
2008/0264925	12/2007	Lockhart	N/A	N/A
2009/0049764	12/2008	McCulloch et al.	N/A	N/A
2009/0211148	12/2008	McCarty	N/A	N/A
2009/0260276	12/2008	Kirsch et al.	N/A	N/A
2010/0065094	12/2009	Ways	N/A	N/A
2010/0071258	12/2009	Molnar et al.	N/A	N/A
2010/0186792	12/2009	Imhof	N/A	N/A
2010/0329649	12/2009	Potter	N/A	N/A
2011/0063798	12/2010	Denter et al.	N/A	N/A
2011/0289825	12/2010	James	N/A	N/A
2011/0308139	12/2010	James	43/132.1	A01M 1/2094
2012/0060407	12/2011	Lindsey	N/A	N/A
2012/0096761	12/2011	Smith	N/A	N/A
2012/0180381	12/2011	DeMonte et al.	N/A	N/A
2012/0186138	12/2011	Bell et al.	N/A	N/A
2012/0186140	12/2011	Raud et al.	N/A	N/A
2012/0192479	12/2011	Schmitz	43/132.1	A01M 1/2094
2012/0216444	12/2011	Raud et al.	N/A	N/A
2012/0233907	12/2011	Pattison et al.	N/A	N/A
2013/0263496	12/2012	Maloney et al.	N/A	N/A
2013/0269239	12/2012	Whitley et al.	N/A	N/A
2013/0276357	12/2012	Shannon et al.	N/A	N/A
2013/0276358	12/2012	Knote et al.	N/A	N/A
2014/0013653	12/2013	Lander	N/A	N/A
2018/0360019	12/2017	Pattison	N/A	A01M 1/2094

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2387235	12/2001	CA	N/A
2876471	12/2014	CA	A01M 1/20
43 40 798	12/1994	DE	N/A
692 30 043	12/1999	DE	N/A
202004003142	12/2003	DE	N/A
698 36 759	12/2006	DE	N/A
603 16 443	12/2007	DE	N/A
0 298 599	12/1988	EP	N/A

0 575 952	12/1992	EP	N/A
0 673 387	12/1998	EP	N/A
0 963 694	12/1998	EP	N/A
0 963 694	12/2001	EP	N/A
1 220 604	12/2003	EP	N/A
1 029 044	12/2005	EP	N/A
1 034 273	12/2005	EP	N/A
1 356 730	12/2006	EP	N/A
1 811 224	12/2006	EP	N/A
1 521 524	12/2006	EP	N/A
1 942 300	12/2007	EP	N/A
1 706 033	12/2009	EP	N/A
1 443 958	12/2009	EP	N/A
1 283 716	12/2011	EP	N/A
1 571 911	12/2011	EP	N/A
359938	12/1930	GB	N/A
1092421	12/1966	GB	N/A
10089853	12/1966	GB	N/A
1 244 414	12/1970	GB	N/A
2 096 569	12/1981	GB	N/A
2 195 875	12/1986	GB	N/A
2 195 078	12/1987	GB	N/A
2 206 790	12/1988	GB	N/A
2430158	12/2006	GB	N/A
57-75654	12/1981	JP	N/A
58-72504	12/1982	JP	N/A
59-196439	12/1983	JP	N/A
59-196440	12/1983	JP	N/A
60-61509	12/1984	JP	N/A
60-199804	12/1984	JP	N/A
64-37958	12/1988	JP	N/A
64-38004	12/1988	JP	N/A
1-93505	12/1988	JP	N/A
1-156901	12/1988	JP	N/A
1-170468	12/1988	JP	N/A
1-212560	12/1988	JP	N/A
1-282619	12/1988	JP	N/A
2-40306	12/1989	JP	N/A
2-65868	12/1989	JP	N/A
2-282308	12/1989	JP	N/A
3-44305	12/1990	JP	N/A
3-133906	12/1990	JP	N/A
3-151308	12/1990	JP	N/A
4-185766	12/1991	JP	N/A
4-185767	12/1991	JP	N/A
4-217606	12/1991	JP	N/A
4-235141	12/1991	JP	N/A
4-343853	12/1991	JP	N/A
5-4901	12/1992	JP	N/A
5-17310	12/1992	JP	N/A

1096215	12/1992	JP	N/A
5-178706	12/1992	JP	N/A
5-255026	12/1992	JP	N/A
5-294895	12/1992	JP	N/A
6-16503	12/1993	JP	N/A
6-311980	12/1993	JP	N/A
6-346373	12/1993	JP	N/A
7-173005	12/1994	JP	N/A
7-279049	12/1994	JP	N/A
7-279050	12/1994	JP	N/A
7-287013	12/1994	JP	N/A
7-316003	12/1994	JP	N/A
8-56543	12/1995	JP	N/A
8-188510	12/1995	JP	N/A
8-188966	12/1995	JP	N/A
9-67396	12/1996	JP	N/A
9-108337	12/1996	JP	N/A
9-159584	12/1996	JP	N/A
9-227305	12/1996	JP	N/A
10-36201	12/1997	JP	N/A
10-150902	12/1997	JP	N/A
10-203907	12/1997	JP	N/A
11-19088	12/1998	JP	N/A
11-240802	12/1998	JP	N/A
2001-120145	12/2000	JP	N/A
2001-157538	12/2000	JP	N/A
2001-316212	12/2000	JP	N/A
2003-95827	12/2002	JP	N/A
2004-65140	12/2003	JP	N/A
2005-255209	12/2004	JP	N/A
WO 94/13698	12/1993	WO	N/A
WO 97/15600	12/1996	WO	N/A
WO 98/16107	12/1997	WO	N/A
WO 99/24567	12/1998	WO	N/A
WO 99/27104	12/1998	WO	N/A
WO 01/80881	12/2000	WO	N/A
WO 01/089295	12/2000	WO	N/A
WO 2001089295	12/2000	WO	N/A
WO 02/05644	12/2001	WO	N/A
WO 03/030931	12/2002	WO	N/A
WO 03/103395	12/2002	WO	N/A
WO 2004/014134	12/2003	WO	N/A
WO 2004/045291	12/2003	WO	N/A
WO 2004/082376	12/2003	WO	N/A
WO 2005/067798	12/2004	WO	N/A
WO 2006/188503	12/2005	WO	N/A
WO 2007/027601	12/2006	WO	N/A
WO 2008/051501	12/2007	WO	N/A
WO 2010/036331	12/2009	WO	N/A
WO 2010/096184	12/2009	WO	N/A

WO-2017070002	12/2016	WO	N/A
WO-2019191021	12/2018	WO	N/A
WO-2021005612	12/2020	WO	N/A

OTHER PUBLICATIONS

“Bed Bug: Heat Treatment Equipment and Application Training” literature, Temp-Air, undated, 2 pages. cited by applicant

Chromalox, Inc. brochure, “ThermEx Model TRMX Heat Remediation Solution,” undated, 1 page. cited by applicant

Fantech, “Installation and Maintenance Instructions for Model FG,” Jul. 18, 2007 and Apr. 7, 2010, 10 pages. cited by applicant

“Insect Heat Treatment—Food Processing Plants, Warehouses & Bins/Silos” brochure, Temp-Air, undated, 2 pages. cited by applicant

International Search Report and Written Opinion mailed Sep. 10, 2012. cited by applicant

Kells, Stephen, “Key Findings on Bed Bug Behavior at Lethal & Sub-lethal Temperatures,” Department of Entomology, University of Minnesota, undated, 2 pages. cited by applicant

Leventhal, Stephanie, “UF creates cheap way to battle bed bugs,” article from The Independent Florida Alligator, Jul. 14, 2009, 3 pages. cited by applicant

Lupo, Lisa, “Treating with Heat, A Simple Principle—A Complex Application,” QA Magazine, Mar./Apr. 2008, 4 pages. cited by applicant

“NorpacR2 Arctic Outdoor Shelter Fabric” press release, dated prior to Aug. 2007 ice fishing season, 2 pages. cited by applicant

Pereira et al., “Bed Bugs: Containerized fumigation and heat treatment,” University of Florida, 100 pages (Sep. 2008). cited by applicant

Pereira et al., “Lethal Effects of Heat and Use of Localized Heat Treatment for Control of Bed Bug Infestations,” Journal of Economic Entomology, vol. 102, No. 3, Entomological Society of America, Jun. 2009, 7 pages. cited by applicant

Pereira et al., University of Florida, www.pctonline.com, pp. 34-39 (Mar. 2009). cited by applicant

“Pest Control” product literature, TempAir, undated, 12 pages. cited by applicant

Printout from http://www.bedbugchaser.com/about_BedBug_Chaser.html, dated Jun. 14, 2011, 1 page. cited by applicant

Printout from <http://www.mypmp.net/bed-bugs/the-heat-on?print=1> printed Dec. 4, 2009 from a post dated Nov. 23, 2009, 1 page. cited by applicant

“Propane-fueled Thermal Remediation” technology fact sheet, Propane Education and Research Council, Mar. 2007, 2 pages. cited by applicant

Smith, Charles, “Turning the heat up under infested homes/popular alternative to fumigation attacks, mold and fungus,” posted on SFGate.com, dated Aug. 30, 2003, accessed via http://articles.sfgate.com/2003-08-30/home-and-garden/17505027_1_heat-treatment-fumi... On Dec. 15, 2009, 2 pages. cited by applicant

Tjernlund AirShare Ventilators products brochure, Jan. 2009, 2 pages. cited by applicant

Tjernlund UnderAire Crawl Space Ventilators products brochure, 2006, 2 pages. cited by applicant

Tjernlund Ventilation Products brochure, copyright 2009, 4 pages. cited by applicant

Potter, “Bed Bugs,” UK Cooperative Extension Service, pp. 1-14 (Revised Jul. 2020). cited by applicant

Primary Examiner: Ellis; Christopher P

Attorney, Agent or Firm: Merchant & Gould P.C.

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 18/348,754, filed Jul. 7, 2023, now U.S. Pat. No. 12,063,921, issued Aug. 20, 2024, which is a continuation of U.S. application Ser. No. 17/236,187, filed Apr. 21, 2021 now U.S. Pat. No. 11,737,445, issued on Aug. 29, 2023, which is a continuation of U.S. application Ser. No. 16/050,016, filed Jul. 31, 2018, now U.S. Pat. No. 11,013,226, issued May 25, 2021, which is a divisional of Ser. No. 14/954,352, filed Nov. 30, 2015, now U.S. Pat. No. 10,070,639, issued Sep. 11, 2018, which is a divisional of U.S. application Ser. No. 13/421,409, filed Mar. 15, 2012, now U.S. Pat. No. 9,226,489, issued Jan. 5, 2016, which claims priority to Provisional Application Ser. No. 61/454,255 filed Mar. 18, 2011, which are both incorporated by reference herein in their entirety.

FIELD

(1) The present disclosure generally relates to the field of pest elimination. The present disclosure includes articles, systems, and methods of heat treatment to target and kill pests such as arthropods, rats, and mice. The present disclosure is especially suited for killing all life stages of bed bugs.

BACKGROUND

(2) Some methods of pest treatment require discarding suspected infested articles. These items can be expensive to replace, especially in the case of large objects like furniture, mattresses, and box springs. Some methods of pest treatment exclusively rely on pesticides, which may be restricted or prohibited on certain items, particularly items that contact people. Finally, some methods of pest treatment use heat but can damage objects, create cold spots that provide a harborage for pests, require large amounts of heat, or are energy inefficient, for example in the case of heat treatments that heat entire rooms or buildings.

(3) It is against this background that the present disclosure is made.

SUMMARY

(4) Surprisingly, it has been found that an infested article or suspected article can be treated by using high heat temperatures and without damaging the article itself.

(5) In some embodiments, the disclosure relates to a method of treating an article suspected of being infested. In the method, an article is placed inside of an enclosure, the enclosure is sealed, and the temperature inside of the enclosure is ramped up at a rate of from about 10 to about 15 degrees Fahrenheit per hour, or 1 degree Fahrenheit every six minutes, until a temperature of at least 115° F. is reached. Once 115° F. is reached, the temperature is held for at least four hours. Thereafter, in some embodiments, the temperature may be slowly decreased at a rate of about 10 to about 15 degrees an hour, or one degree every six minutes, until the original starting temperature (which may be room temperature) is reached.

(6) In some embodiments, the disclosure relates to a method of automatically adjusting the temperature inside of a sealed enclosure in order to treat an article. In this method, a flexible, inflatable enclosure and heating system are provided. The heating system has a heater. The heating system also has at least one thermocouple electrically connected to the heater and located inside of the inflated enclosure. The thermocouple measures the actual air temperature inside of the enclosure ($T_{sub,actual}$). The heating system also includes a programmable logic controller electrically connected to the heater and the thermocouple and programmed with a predetermined temperature ramp rate controlled by a set point temperature ($T_{sub,setpoint}$) and a target temperature ($T_{sub,target}$). The programmable logic controller increases the temperature inside of the enclosure by increasing the $T_{sub,setpoint}$ one degree every six minutes until $T_{sub,target}$ is reached, comparing the $T_{sub,actual}$ to the $T_{sub,setpoint}$, and adjusting the heater in response to the difference between $T_{sub,actual}$ and $T_{sub,setpoint}$. In some embodiments, the programmable

logic controller also controls the decrease in temperature.

(7) In some embodiments, the disclosure relates to a flexible and inflatable enclosure for treating an article suspected of being infested with an arthropod or other pest. The enclosure has a floor, a ceiling, and at least one wall, where at least part of the ceiling or the wall can be partially removed to allow for the selected article to be placed inside of the enclosure.

(8) In some embodiments, the disclosure relates to a treatment system. The system includes a flexible and inflatable enclosure having a floor, a ceiling, and at least one wall, where at least part of the ceiling or wall can be at least partially removed to allow for the selected article to be placed inside of the enclosure. The system also includes a heating system with a first heater, a thermocouple electrically connected to the heater and configured to be located inside of the inflated enclosure, and a programmable logic controller electrically connected to the first heater and the thermocouple. In some embodiments, the system can also include a fan designed to be used inside of the inflated enclosure, and an external support structure.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) FIG. 1 shows a diagram of an exemplary enclosure and heating system.

(2) FIG. 2 shows a diagram of an exemplary enclosure and heating system.

(3) FIG. 3 shows a diagram of an exemplary enclosure and heating system.

(4) FIG. 4 shows a schematic of an exemplary enclosure and heating system.

(5) FIG. 5 shows the exterior of a heater.

(6) FIG. 6 shows the interior of a heater.

(7) FIG. 7 shows temperature data from a hotel room heat test.

(8) FIGS. 8A and 8B show bed bug adult and egg mortality at various temperatures.

(9) FIG. 9 shows temperature data from a heated enclosure with a mattress in it.

(10) FIG. 10 shows a diagram of an exemplary temperature control program for the programmable logic controller.

(11) In accordance with common practice, the various described features are not drawn to scale but are drawn to emphasize certain features relevant to the present disclosure. Reference characters denote like features throughout the Figures.

DETAILED DESCRIPTION

(12) In some embodiments, the present disclosure relates to articles, systems, and methods of heat treatment to target and kill pests including all life stages of bed bugs. Surprisingly, it has been found that gradually increasing the temperature inside of an enclosure allows for heat treatment of articles that are believed to be infested with pests without adversely affecting the article. In the past, these articles may have been discarded instead of treated in part because their size and internal spaces did not lend them to be effectively treated with pesticides, traditional heat treatments can damage the article, and label restrictions on pesticides did not allow for treatment of items that contact humans on a regular basis. Exemplary articles include mattresses, box springs, and furniture, although any article would benefit from the disclosed methods, articles, and systems. Articles, systems, and methods disclosed herein are effective at eliminating pests including 100% of bed bug eggs, nymphs, and adults.

(13) The Enclosure and Treatment System

(14) In some embodiments, the present disclosure includes a flexible and inflatable enclosure. Using a flexible and inflatable enclosure allows the enclosure to be easily set up and taken down from one treatment site to the next. The enclosure, when inflated, can take on a variety of shapes including a sphere, a cylinder, a cube, and a box. A cube and a box are preferred shapes because they are practical and easy to manufacture. A sphere with a flat bottom is also a preferred shape

because it circulates air uniformly throughout the interior which minimizes cold spots. While an inflatable enclosure is disclosed, it is understood that other enclosures could be used such as a flexible enclosure with an internal or external frame to hold the enclosure up. In some embodiments, the enclosure does not need to be inflatable or flexible. In these embodiments, the enclosure may be rigid. The rigid enclosure may also be foldable or collapsible. Exemplary rigid enclosures could include pre-formed rigid panels that are assembled into the final enclosure, foldable panels that can be assembled into the final enclosure, or expandable panels that can be assembled into the final enclosure. The rigid enclosure may optionally be insulated.

(15) An exemplary enclosure is shown in FIG. 1. Alternative enclosures are shown in FIGS. 2, 3, and 4. Referring now to FIG. 1, if the enclosure forms a cube or a box, then the enclosure **10** includes a floor **12**, a ceiling **14**, and four walls **16**, **18**, **20**, and **22**. While four walls are shown in FIG. 1, it is understood that shapes with one wall, or other combination of walls, could be used (as in the case of a sphere or cylinder). Note that the wall does not need to be flat. In order to facilitate getting large objects in and out of the enclosure, one wall **18** and the ceiling **14** are preferably at least partially removable. In some embodiments, the one wall **18** and the ceiling **14** can be connected to each other, and the ceiling can be connected to the rest of the enclosure on one side (near wall **22** for example), so that the ceiling and the wall together form a large flap that comes over the top and down one side of the enclosure. This “flap” can be connected to the rest of the enclosure, for example by using a zipper **24**. In other words, the enclosure can be opened by unzipping one wall and the ceiling, and peeling back the wall and ceiling to allow for an article to be placed inside of the enclosure, and then the ceiling and the wall can be placed back into position and zipped up to seal the enclosure. In some embodiments, the enclosure includes a hook and loop sealing system that goes over the zipper to further insulate the interior of the enclosure. In some embodiments, the enclosure may also include a hook and loop sealing system on the exterior of each window so as to minimize any heat loss when the window is not being used.

(16) In some embodiments, the enclosure includes an external support structure, skeleton, or frame **26**. While such a structure, skeleton, or frame may be unnecessary in order for the enclosure to be able to inflate and stand on its own, it has been found that a structure, skeleton, or frame aids in preventing the formation of cold spots on the inside of the enclosure. If part of the enclosure collapses or a corner gets pinched while the enclosure is inflated, then the hot air may not circulate to that portion of the enclosure. If the temperature of that spot does not rise to 115° F. and remain there for the designated period of time, that spot may become a harborage for pests. If pests survive inside the enclosure, then the enclosure can actually become a source of contamination when the enclosure is taken down and transported to the next location. In some embodiments, the structure is a plurality of bars located on the corners of the enclosure. In some embodiments, the structure is a scaffolding or frame that attaches to the bottom, sides, and top of the enclosure.

(17) In some embodiments, the enclosure is made of insulated material that makes the enclosure energy efficient. An example of insulated material is the Norpac R2 and Norpac M3 Insulation, commercially available from Norpac LLC (Rogers, MN), which may be used together as a combination Norpac R2/M3 material from Norpac LLC and can also be used together with Thinsulate™ Insulation FR from 3M (St. Paul, MN). Other types of insulation that might be suitable include Thermal-Wrap® nuclear-grade blanket insulation (Transco Products), HeatShield products (Heatshield), heat blankets (Heatcon), swimming pool thermal covers, outdoor plant covers, Kevlar tarps, polystyrene, Thinsulate, insulated canvas, vinyl, space blankets, Nomex, and Thermarest materials. In some embodiments, the material selected has an R-value, which is a measure of thermal resistance, of about 0.5, about 0.8, about 1, about 1.4, about 2, about 3, about 4, at least 0.5, at least 0.8, at least 1, at least 1.4, at least 2, at least 3, or at least 4. A person skilled in the art will understand that achieving the desired temperatures is a combination of temperature, insulation, time and energy variables. Increasing the insulation may mean that the energy going into the system can be reduced. Likewise, decreasing the insulation may mean that the energy

going into the system needs to be increased. The insulation material can be modified depending on whether the enclosure is designed to be reused or a one-time use enclosure. In some embodiments, the enclosure is made from flame retardant material. In some embodiments, the enclosure is made from a material that does not absorb pesticides and can be cleaned easily. In some embodiments, the enclosure includes a layer of canvas, vinyl, or other thick, durable material on the floor that is designed to protect the floor from the articles that are moved in and out of the enclosure. In some embodiments, the interior of the enclosure is made of a light colored material. Using a light colored material allows users or operators to visually inspect the inside for pests. In some embodiments, the enclosure is made from material that is configured for a one-time use that is disposed of after that use. In other embodiments, the enclosure is made from material that is configured to be reused.

(18) In some embodiments, the enclosure has one or more windows **28** located in at least one wall. In some embodiments, the enclosure includes a window on each wall. In some embodiments, the enclosure includes an inlet hole **30** and an outlet hole **32**. When the enclosure is connected to a heating system, the duct **34** bringing air from the heater is placed in the inlet hole **30** and the duct **36** removing air from the enclosure is placed in the outlet hole **32**. The enclosure can also include an optional thermocouple **52** and thermocouple securing device **54** located in the cold spot of the enclosure. The location of the cold spot varies depending on the heater location and article setup. For example, when the heater is in a corner of a cube, then the cold spot is usually in the corner that is diagonally opposite the heater. When the heater is placed in the middle of a wall of a cube or rectangle, then the cold spot may be in both corners of the opposite wall. Further, the articles that are located in the enclosure can create cold spots. For example, if bed linens are piled in a corner without airflow under or around it, a cold spot can form underneath the linens. By placing a thermocouple **52** in the cold spot, the operator can ensure that the temperature of the cold spot reaches 115° F. and maintains that temperature over time. Exemplary thermocouple securing devices **54** include a strap secured by snaps or hook and loop, a buckle, or a permanent sleeve or loop that the thermocouple is threaded through.

(19) In some embodiments, the floor is connected to the walls with a seam where the seam is at least one inch off of the ground. The advantage of having the seam slightly off of the ground is that it makes it less likely that pests will find the seam and nest in it.

(20) When the enclosure is inflated, it preferably has an internal volume of at least about 6, about 50, about 368, or about 800 cubic feet.

(21) Referring now to FIGS. **5** and **6**, the enclosure is used in conjunction with a heating system **38**. The heating system includes a first heating element **40**, at least one thermocouple **52** attached to the thermocouple plug **42** and electrically connected to the heater and configured to be located inside of the inflated enclosure, and a programmable logic controller **44** electrically connected to the first heater and the thermocouple. It is understood that a PC board or other controller could be used instead of a programmable logic controller. The heating system can also include an internal fan **50**. The heater may include a split inlet duct that allows either fresh air or return air to be run through the heater. Alternatively, the duct may not be split, but may include a manifold that allows for either fresh or recirculated air to flow into the enclosure. Using return air may be more energy efficient because the air is already hotter than ambient air. The heating system is connected to the enclosure through the ducts **34** and **36**, shown in FIG. **1**. In some embodiments, duct **34** may include a manifold that opens and closes to the outside. This manifold would allow fresh air to circulate into the enclosure. This is useful and can be open when inflating the enclosure. Once the enclosure is inflated, it may be beneficial to close the fresh air manifold and allow only recirculated hot air to circulate. This would avoid having to heat fresh air. Duct **36** may also include a manifold that opens and closes to the inside of the enclosure. This manifold may be closed while the enclosure is inflating. Once the enclosure is inflated, the manifold can be opened to allow hot air to recirculate from the enclosure to the heater and then back to the enclosure. In some embodiments, the heating system **38**, instead of duct **34**, may include the manifold that opens to the outside. The heating

system also includes a fan **50** for blowing the hot air into the enclosure and for facilitating the return of air from inside of the enclosure. The fan **50** is also responsible for inflating the enclosure. In some embodiments, the heater includes a second heating element **46** to maximize heat output. If two heating elements are used, then the system preferably works off of two different 15 amp circuits. In one exemplary embodiment, the first heating element can be sized to accommodate a maximum 12 amp current draw on the circuit while powering the heating element and the fan internal to heater and the second heating element is slightly larger and accommodates the PLC while staying under the maximum 12 amp current draw.

(22) The heater can be equipped with various safety features. For example, the controller can be programmed to automatically start the shut down process or shut off completely if the temperature does not reach a desired temperature within a longer period of time, or if it reaches too hot of a temperature within a period of time. For example, if the temperature does not reach a predetermined temperature such as 110° F. or 115° F. within a certain period of time such as 10 hours, 12 hours, or 16 hours, the heater can start decreasing the temperature by 1 degree every 6 minutes or 5 or 10 degrees every hour. In another example, if the temperature reaches a higher temperature such as 120° F., 125° F., or 130° F. within an hour, 2 hours, or 4 hours, the controller can be programmed to start the process of decreasing the temperature by 1 degree every 6 minutes or 5 or 10 degrees every hour. Also, in this example, the controller can be programmed to simply shut down the heater altogether and allow the temperature to cool down on its own. The controller can also be programmed to alert a user or operator that the treatment was not successful. This would allow the user or operator to correct any problems with the system and run the cycle again, or provide an alternative pest treatment. The controller can also be equipped to remotely alert the user or operator that there is a problem on the user or operator's mobile phone, pager, email, or text.

(23) In addition to the enclosure and the heating system, the system can optionally include an internal fan inside of the inflated enclosure. This fan helps circulate the air inside of the enclosure to reduce the likelihood of cold spots inside of the enclosure. Eliminating cold spots inside of the enclosure is important. If the temperature of any spot inside of the enclosure does not reach the minimum temperature needed to kill the pests, then that spot can become a source of contamination. If the system includes an internal fan, the system preferably includes relays and controls that determine if the internal fan inside of the enclosure was powered on and working throughout the heating cycle. The system can also include a stand for supporting the article to be treated. The stand should be designed in a way to maximize the air circulation around the article while minimizing cold spots in and around the article and around the stand. In some embodiments, the stand is L-shaped. In some embodiments, the stand has a minimum of a 1-inch clearance between the floor and the article on the stand to ensure proper circulation. The stand is preferably made of a heat-conducting material such as metal. This also helps eliminate cold spots.

(24) Methods of Use

(25) To set up the enclosure, the deflated enclosure is spread out and hooked up to the heater. The articles are placed inside of the deflated enclosure. In some embodiments, the items are placed in the enclosure in such a way that the item has at least a 1 inch space on all sides between the article and the floor, walls, and other articles in the enclosure once the enclosure is set up or inflated. In some embodiments, the article has at least a 1-2 inch space around the article and the walls, floors, or other articles of the assembled or inflated enclosure. Leaving space around the article is important to assure proper air flow around the article. The heater is turned on and the manifold in the inlet duct **34** is opened to allow air to flow into the deflated enclosure in order to heat and inflate it. While the enclosure is inflating, the manifold in outlet duct **36** is closed. Once the enclosure is inflated, the manifold in the inlet duct **34** is closed, and the manifold in outlet duct **36** is opened. This allows for the warm air inside of the enclosure to be recirculated.

(26) In some embodiments, the present disclosure relates to a method for treating a suspected infested article. The method includes placing an article inside of an enclosure such as the one

described above, sealing the enclosure, and then adjusting the temperature inside of the sealed enclosure. In a preferred embodiment, the temperature is gradually increased at a predetermined controlled rate, and then held at a peak temperature for a predetermined period of time, and then gradually decreased at a predetermined controlled rate. In an embodiment, the temperature is increased from room temperature at a rate of from about 10 to about 15 degrees Fahrenheit every hour or one degree every six minutes. The temperature may be increased more slowly over a longer period of time. For example, the temperature can be ramped up at from about 1 to about 10 degrees Fahrenheit every hour. But, the temperature should not be ramped up too quickly for sensitive articles or the article may be damaged. The temperature is increased until the temperature reaches a predetermined peak temperature. The various life stages of bed bugs die at a temperature of about 115 degrees Fahrenheit. But, the articles being treated can include larger objects that have internal spaces that could be infested with pests (i.e., furniture, mattresses, and box springs). Therefore, it may be beneficial to increase the temperature higher than the minimum needed to kill all life stages of pests. Therefore, in some embodiments, the temperature inside of the enclosure is increased to at least about 115° F., at least about 118° F., at least about 120° F., at least about 130° F., or at least about 180° F. and then held at that temperature for a predetermined period of time. The predetermined temperature hold time will vary depending on the temperature, but the temperature can be held from about 1 minute to about 10 hours, about 1 hour to about 5 hours, about 2 hours to about 4 hours, or longer periods of time such as about 2 hours to about 8 hours, or about 8 hours to about 24 hours. After the predetermined period of time has elapsed, the temperature inside of the enclosure is gradually ramped down from the peak temperature to room temperature at a predetermined controlled rate. In an embodiment, the temperature is decreased to room temperature at a rate of from about 10 to about 15 degrees Fahrenheit every hour. As with the ramp up, the temperature can be decreased more slowly or over a longer period of time. Care should be taken not to decrease the temperature too quickly for sensitive articles and risk damaging the articles. Alternatively, the heater can be turned off and the temperature inside of the enclosure can be allowed to return to room temperature on its own. This is most effective when the insulation on the enclosure helps the temperature decrease slowly enough, or where the articles inside of the enclosure are not sensitive to a rapid temperature decrease.

(27) It is especially important to control the temperature increase and decrease when treating articles that are made of multiple materials, such as furniture. Furniture often is made of multiple types of wood, composites, laminates, plastics, and adhesives. If the temperature of the furniture is increased or decreased too quickly, the different materials may change temperature at different rates which could cause the piece of furniture to warp. An alternative embodiment of the present disclosure can be used with articles that are not as temperature sensitive as furniture where the temperature is increased or decreased as quickly as possible. This can be useful when treating non-sensitive items such as soft goods, bedding, curtains, towels, personal items, and clothes. In one preferred embodiment, the disclosed system is designed to have two modes of operation where a user can select a rapid temperature increase and decrease if non-temperature-sensitive articles are being treated or a controlled, slow temperature increase and decrease if temperature sensitive articles are being treated.

(28) In some embodiments, the temperature is automatically adjusted by the programmable logic controller on the heating system. In this method, the heating system has a heater. The heating system also has at least one thermocouple electrically connected to the heater and located inside of the inflated enclosure. The thermocouple measures the actual air temperature inside of the enclosure ($T_{sub,actual}$). The heating system also includes a programmable logic controller electrically connected to the heater and the thermocouple. In some embodiments, the programmable logic controller is programmed with predetermined temperature ramp rate controlled by a set point temperature ($T_{sub,setpoint}$) and a target temperature ($T_{sub,target}$). In some embodiments, the controller can be programmed by a user or operator to vary the temperature ramp

rate and hold time. An exemplary predetermined ramp rate may be 1 degree Fahrenheit every six minutes. It is understood that other ramp rates could be used including partial degree increments. It is also understood that step increases could be used where the programmable logic controller increases the temperature by, for example, 5 or 10 degrees and then holds that temperature for a period of time such as 60 minutes. In one exemplary embodiment, the programmable logic controller increases the air temperature inside of the enclosure by increasing the T.sub.setpoint one degree every six minutes until T.sub.target is reached, comparing the T.sub.actual to the T.sub.setpoint, and adjusting the heater in response to the difference between T.sub.actual and T.sub.setpoint. The heater can be adjusted in several ways. For example, the heater can be adjusted by simply turning the heater on and off. The heater can also be adjusted by increasing or decreasing the power to the heater. And the heater can be adjusted by modifying a ratio of "on time" to "off time". If $T_{sub.actual} < T_{sub.setpoint}$, then the programmable logic controller can turn the heater on, increase the power to the heater, increase the amount of "on time," or decrease the amount of "off time." If $T_{sub.actual} > T_{sub.setpoint}$, then the programmable logic controller can turn the heater off, decrease the power to the heater, decrease the amount of "on time," or increase the amount of "off time." Once $T_{sub.setpoint} = T_{sub.target}$, the programmable logic controller holds the T.sub.target for a predetermined period of time (for example, for four hours) and continues to compare the T.sub.actual to the T.sub.setpoint (which is now also T.sub.target) and adjust the heat up or down to maintain T.sub.target. After the temperature has been held for the predetermined period of time, in some embodiments, the programmable logic controller also controls the decrease in air temperature inside of the enclosure by decreasing the T.sub.setpoint one degree every six minutes until $T_{sub.actual} =$ the original T.sub.actual, which could be room temperature, comparing the T.sub.actual to the T.sub.setpoint, and turning the heater on or off in response to the difference between T.sub.actual and T.sub.setpoint. If $T_{sub.actual} < T_{sub.setpoint}$, then the programmable logic controller can turn the heater on, increase the power to the heater, increase the amount of "on time," or decrease the amount of "off time." If $T_{sub.actual} > T_{sub.setpoint}$, then the programmable logic controller can turn the heater off, decrease the power to the heater, decrease the amount of "on time," or increase the amount of "off time." This is generally shown in FIG. 10.

(29) In an embodiment, the heating system can include a second thermocouple where a first thermocouple measures the air temperature inside of the enclosure and a second thermocouple measures the air temperature inside of an article, such as a piece of furniture or a mattress. In this embodiment, the first thermocouple controls the temperature ramp rate controlled by a set point temperature, T.sub.setpoint, (both increase and decrease) but the second thermocouple controls whether the temperature inside of the enclosure has reached T.sub.target. Using two thermocouples is a more sophisticated way to control the temperature increase and decrease while ensuring that the air temperature inside of an article maintains a high enough temperature.

(30) Exemplary articles to be treated include a mattress, a boxspring, bedding, nightstands, furniture, mirrors, pictures, light fixtures, window treatments, clothing, appliances, commercial fixtures, telephones, remote controls, alarm clocks and small electrical items, luggage, personal items, and pet bedding. The article can be located in a variety of places including a hotel, a house, an apartment or multi-family complex, restaurants, an office building, a movie theater, a train, a bus, an airplane, a car, a truck, a retail store, a college dormitory, a doctor's office, a veterinary clinic, a hospital, and a nursing home. The article can also be a large article such as manufacturing equipment, a bus, a car, a truck, an airplane, or a train.

(31) In addition to using heat to kill pests, the methods disclosed herein can be used in conjunction with pesticides and/or as part of a pest treatment program.

(32) The present disclosure may be better understood with reference to the following examples. These examples are intended to be representative of specific embodiments of the disclosure, and are not intended as limiting the scope of the disclosure.

EXAMPLES

Example 1—Hotel Room Heat Test

(33) FIG. 7 shows temperature data collected during a treatment of an entire hotel room. The entire room was heated up and the temperature was taken at different locations within the room. The lethal temperature needed to kill bed bugs is 115° F. FIG. 7 shows that the ceiling of the room gets very warm, but the floor is still below the lethal temperature, making it a place for bed bugs to hide and survive the treatment. Lethal temperatures may be achieved at the floor but large fans and substantially more electric power would be required which could be costly and not practical.

Example 2—Efficacy of High Temperatures on Bed Bug Adults and Eggs Over Time

(34) Example 2 evaluated the temperature required to kill bed bug adults and eggs. The example evaluated three elevated temperatures (100° F., 115° F., and 130° F.) for mortality among bed bug adults and eggs when exposed to heat for 30 minutes, 1 hour, 2 hours and 4 hours. Ten adult bed bugs and ten eggs were evaluated for each time and temperature period. The eggs were observed at two weeks for hatch. The bed bug adults were evaluated immediately upon removal from the heated container. FIGS. 8A and 8B show the efficacy of high temperatures on bed bug adults and eggs over time. The figures show that that temperature alone is not enough to be effective, but that a combination of high temperatures and time is needed. FIGS. 8A and 8B show that maintaining 115° F. for four hours is effective against eggs and adults. FIGS. 8A and 8B also show that if the temperature is increased, the amount of time can be decreased as 130° F. was effective against eggs and adults in 30 minutes.

Example 3—Heat Trial Run on a Mattress

(35) Example 3 determined the heat penetration into the interior of a mattress. For this example, thermocouples were placed at various locations inside of an enclosure and inside of a mattress. FIG. 9 shows that an internal air temperature inside of the enclosure of 123° F. is high enough to raise the temperature of the interior of the mattress to the required 115° F. The air around the exterior and the air in the interior of the article being treated must reach 115° F., otherwise, the interior spaces of the article, particularly with large articles like furniture and mattresses, can become a harborage for bed bugs.

(36) The above specification, examples and data provide a complete description of the disclosed compositions and methods of use. Since many embodiments of the disclosure can be made without departing from the spirit and scope of the invention, the invention resides in the claims.

Claims

1. A pest treatment system comprising: a flexible enclosure comprising: an interior comprising a volume of at least 50 cubic feet when erected; a floor; and at least one side, wherein the at least one side can be at least partially removed to allow for articles to be placed inside of the flexible enclosure; and a heating system comprising: at least one heater; and at least one temperature sensor connected to the heater and configured to be located inside of the flexible enclosure; and an internal frame.
2. The pest treatment system of claim 1, further comprising a fan for the interior of the flexible enclosure.
3. The pest treatment system of claim 1, the flexible enclosure further comprising a ceiling when erected.
4. The pest treatment system of claim 1, further comprising a controller connected to the at least one heater.
5. The pest treatment system of claim 1, wherein the flexible enclosure can be opened and closed with a zipper.
6. The pest treatment system of claim 1, wherein the flexible enclosure can be folded and transported.
7. The pest treatment system of claim 1, wherein the floor is made of canvas or vinyl.

8. The pest treatment system of claim 1, wherein the floor and the at least one side connect at a seam that is at least one inch off a ground.
9. The pest treatment of claim 1, further comprising an inlet hole in the flexible enclosure.
10. A method of treating an article comprising: (a) placing the article inside of a pest treatment system comprising: (i) a flexible enclosure comprising: an interior comprising a volume of at least 50 cubic feet when erected; a floor; and at least one side, wherein the at least one side can be at least partially removed to allow for the article to be placed inside of the flexible enclosure; (ii) a heating system comprising: at least one heater; and at least one temperature sensor connected to the at least one heater and configured to be located inside of the flexible enclosure; and (iii) an internal frame (b) closing the flexible enclosure; (c) connecting the at least one heater to an electrical circuit; (d) heating the interior of the flexible enclosure to a temperature of at least 115° F.; and (e) maintaining an air temperature in the interior of at least 115° F. for a period of time, wherein the interior of the flexible enclosure is constructed to receive the article, the article comprising furniture, a mattress, a box spring, or a combination thereof.
11. The method of claim 10, wherein the period of time is from about 1 hour to about 5 hours.
12. The method of claim 10, wherein the interior of the flexible enclosure is heated to a temperature of 115° F.-130° F.
13. The method of claim 10, wherein the pest treatment system further comprises a fan for the interior of the flexible enclosure.
14. The method of claim 10, wherein the pest treatment system further comprises a controller connected to the heater.
15. The method of claim 10, wherein the flexible enclosure can be opened and closed with a zipper.
16. The method of claim 10, wherein the flexible enclosure can be folded and transported.
17. The method of claim 10, wherein the floor is made of canvas or vinyl.
18. The method of claim 10, wherein the floor and the at least one side connect at a seam that is at least one inch off a ground.
19. The method of claim 10, wherein the pest treatment system further comprises an inlet hole in the flexible enclosure.
20. The method of claim 14, wherein the controller can be programmed to automatically shut down the heater.
21. A pest treatment system comprising: a flexible enclosure comprising: a floor; at least one side, wherein the at least one side can be at least partially removed to allow for articles to be placed inside of the flexible enclosure; and an interior volume of at least 50 cubic feet when erected; a heating system comprising: at least one heater; a controller connected to the at least one heater; and at least one fan; and at least one temperature sensor configured to be located inside of the flexible enclosure.
22. The pest treatment system of claim 21, wherein the at least one temperature sensor is monitored from outside of the flexible enclosure.
23. The pest treatment system of claim 22, wherein the at least one temperature sensor is monitored using the controller.
24. The pest treatment system of claim 21, further comprising a fan for the interior volume of the flexible enclosure.
25. The pest treatment system of claim 21, wherein the flexible enclosure can be opened and closed with a zipper.
26. The pest treatment system of claim 21, wherein the flexible enclosure can be folded and transported.
27. The pest treatment system of claim 21, wherein the floor is made of canvas or vinyl.
28. The pest treatment system of claim 21, wherein the floor and the at least one side connect at a seam that is at least one inch off a ground.
29. The pest treatment system of claim 21, further comprising an inlet hole in the flexible

enclosure.

30. The pest treatment system of claim 21, wherein the flexible enclosure further comprises a ceiling when erected.

31. The pest treatment system of claim 21, further comprising at least one support structure.
